

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Susannal Devapriyam.

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

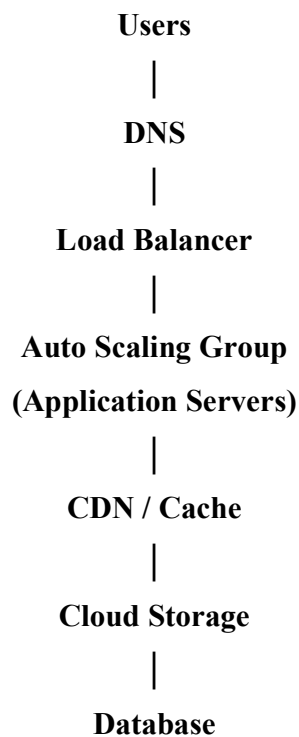
Q1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies.

Answer:

Introduction

A media company that provides online video streaming requires a cloud infrastructure capable of handling millions of users simultaneously. The system should provide fast video delivery, reduce buffering, and remain available even during peak traffic. Cloud computing provides scalable resources, caching, and automatic resource management to achieve these goals.

Proposed Cloud Architecture



Scalability and Caching

- **Scalability** allows the cloud to increase or decrease resources automatically based on user demand.
- **Caching** stores frequently viewed videos in a Content Delivery Network (CDN), reducing buffering and improving loading speed.

Load Balancing and Auto Scaling

- **Load Balancer** distributes user requests equally among multiple servers, preventing overload.
- **Auto Scaling** automatically adds more servers during heavy traffic and removes unused servers during low traffic.

These features ensure smooth video streaming and maintain high performance.

Cost Optimization

The company can reduce costs by:

- Using Auto Scaling to pay only for required resources.
- Storing videos in low-cost cloud storage.
- Using CDN to reduce bandwidth usage.
- Monitoring cloud resources and removing unused services.

Conclusion

By combining scalability, caching, load balancing, and auto scaling, the media company can deliver high-quality video streaming with minimal buffering while optimizing cloud costs.

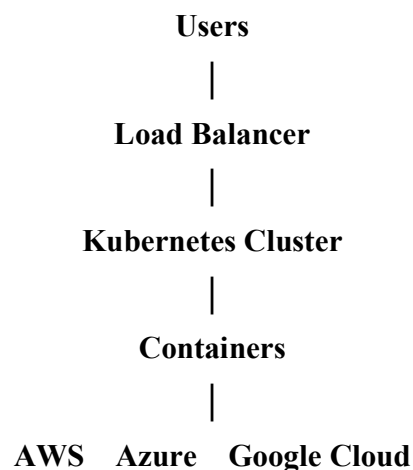
Q2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Answer:

Introduction

A fintech startup needs to deploy secure and reliable applications across different cloud providers. Managing containers manually becomes difficult as the number of applications increases. Container orchestration and multi-cloud strategies help simplify deployment and improve reliability.

Proposed Solution:



Container Orchestration

The startup can use **Kubernetes** to:

- Automatically deploy containers.
- Manage application updates.
- Restart failed containers.
- Balance workload across servers.

Service Discovery and Scaling

- **Service Discovery** allows containers to find and communicate with each other automatically.
- **Auto Scaling** increases or decreases the number of containers based on application traffic.

Benefits of Multi-Cloud

- High availability.
- Reduced vendor dependency.
- Better disaster recovery.
- Improved performance by choosing the best cloud provider.

Risks

- Complex management.
- Higher operational cost.
- Security challenges.
- Difficulty maintaining consistent configurations.

Conclusion

Using Kubernetes with a multi-cloud strategy provides flexibility, scalability, and reliability while reducing dependence on a single cloud provider.

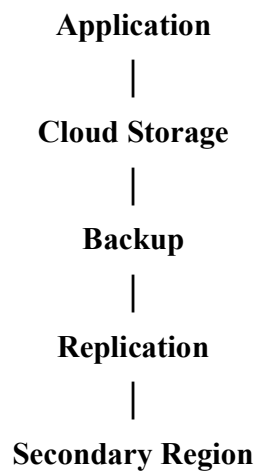
Q3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Answer

Introduction

Data is one of the most valuable resources for an e-learning platform. Accidental deletion or system failures may result in the loss of student records, learning materials, and examination data. A strong backup and disaster recovery plan is essential.

Proposed Solution:



Backup, Replication and Versioning

- **Cloud Backup** creates regular copies of important data.
- **Replication** stores data in another cloud region for safety.
- **Versioning** keeps multiple versions of files, allowing deleted or modified files to be restored.

Achieving RTO and RPO

- **RTO (Recovery Time Objective):** Time required to restore services after failure. Fast backup restoration and replication reduce recovery time.
- **RPO (Recovery Point Objective):** Maximum acceptable amount of data loss. Frequent backups reduce data loss.

Business Continuity

- Perform automatic daily backups.
- Store backups in different cloud regions.
- Regularly test disaster recovery plans.
- Monitor cloud resources continuously.

Conclusion

Using cloud-native backup, replication, and versioning protects important data and allows the organization to recover quickly from failures while maintaining continuous service.

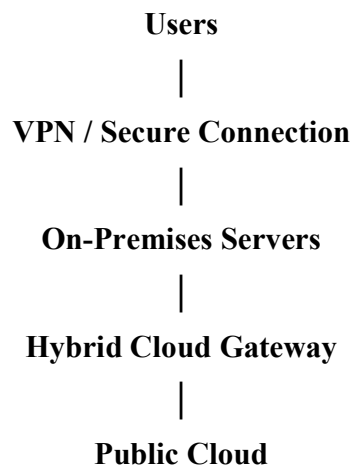
Q4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Answer

Introduction

Government organizations often manage confidential information that cannot be stored completely in the public cloud. A hybrid cloud combines on-premises infrastructure with public cloud services, allowing sensitive data to remain secure while using cloud resources for other applications.

Hybrid Cloud Architecture:



Identity Management

Identity management should include:

- Multi-Factor Authentication (MFA).
- Role-Based Access Control (RBAC).
- Single Sign-On (SSO).

These methods ensure that only authorized users can access sensitive resources.

Encryption

- Encrypt data while storing it (Data at Rest).
- Encrypt data while transferring it (Data in Transit) using SSL/TLS.
- Store encryption keys securely.

Challenges

- Managing two different environments.
- Higher maintenance costs.
- Network latency.
- Security and compliance management.

Conclusion

A hybrid cloud architecture provides flexibility while maintaining security and regulatory compliance for sensitive government data.

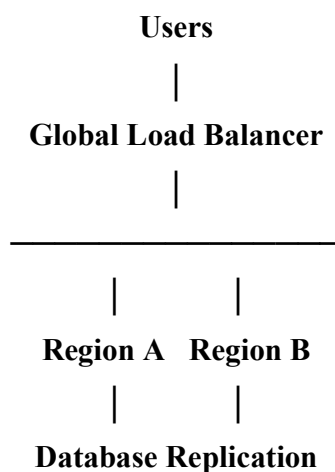
Q5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Answer

Introduction

A SaaS provider must ensure that its application remains available even if one cloud region experiences failure. Multi-region deployment increases reliability and reduces downtime.

Multi-Region Architecture



Data Replication

- Replicate application data between multiple cloud regions.
- Keep backup copies synchronized.
- Ensure that updated data is available in every region.

Failover Mechanism

If Region A fails:

- Traffic is automatically redirected to Region B.
- Users continue using the application without interruption.

Monitoring and Alerting

Cloud monitoring tools continuously check:

- Server health.
- CPU usage.
- Network performance.
- Application availability.

Automatic alerts notify administrators whenever issues occur.

Benefits

- High availability.
- Fault tolerance.
- Reduced downtime.
- Better user experience.

Conclusion

A multi-region deployment with replication, automatic failover, and continuous monitoring ensures reliable cloud services and uninterrupted application availability.